



Agile methods tailoring – A systematic literature review



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ABSTRACT

Background: The software development industry has been adopting agile methods instead of traditional software development methods because they are more flexible and can bring benefits such as handling requirements changes, productivity gains and business alignment. **Objective:** This study seeks to evaluate, synthesize, and present aspects of research on agile methods tailoring including the method tailoring approaches adopted and the criteria used for agile practice selection. **Method:** The method adopted was a Systematic Literature Review (SLR) on studies published from 2002 to 2014. **Results:** 56 out of 783 papers have been identified as describing agile method tailoring approaches. These studies have been identified as case studies regarding the empirical research, as solution proposals regarding the research type, and as evaluation studies regarding the research validation type. Most of the papers used method engineering to implement tailoring and were not specific to any agile method on their scope. **Conclusion:** Most of agile methods tailoring research papers proposed or improved a technique, were implemented as case studies analyzing one case in details and validated their findings using evaluation. Method engineering was the base for tailoring, the approaches are independent of agile method and the main criteria used are internal environment and objectives variables.

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1. Introduction

Agile methods for software development have been increasingly used by the software development industry (Deemer et al., 2010; Nishijima and Dos Santos, 2013; Qumer and Henderson-Sellers, 2006). Compared to traditional software development methods it presents advantages such as accelerate time to market, increase in quality and productivity, improve Information Technology (IT)/business alignment, and enhanced flexibility (Deemer et al., 2010; Jyothi and Rao, 2011; Nishijima and Dos Santos, 2013; Qumer and Henderson-Sellers, 2006; VersionOne, 2013). Even though the benefits of agile methods are worthwhile and have been proved by scientific and market researches (Ahmed and Sidky, 2009; de Azevedo Santos et al., 2011; Glaiel et al., 2013; Jyothi and Rao, 2011; Moniruzzaman and Hossain, 2013; VersionOne, 2013) the complexity of adopting them is high because of organization culture, resistance to change and need for upper management sponsorship and involvement (Chow and Cao, 2008; Dyba and Dingsoyr, 2009; Gandomani et al., 2013; Koch, 2005).

Based on the current challenges of software development, agile methods are interesting and viable options to achieve quality, project budget control, align with organization's business strategy and deliver value frequently and continuously (de Azevedo Santos et al., 2011; Glaiel et al., 2013; Nerur et al., 2005). The strategy choice for agile methods adoption is a key component to get the organization to take advantage of the benefits brought by agility and to overcome the common issues found on the adoption process (Nerur et al., 2005; Soundararajan et al., 2013). The agile adoption process is complex and requires lots of effort from the organization and teams, cultural adaptation, deals with egos and resistance to change, and demands upper management sponsorship (Abbas et al., 2010; Boehm and Turner, 2005; Chow and Cao, 2008; Dyba and Dingsoyr, 2009; Gandomani et al., 2013; VersionOne, 2013). Besides that, there are cases where practices available on the mainstream agile methods don't make sense for an organization (Kurapati et al., 2012). Adopting the practices proposed by an agile method at once means that the organization should need to spend effort and resources adopting it with no evaluation of the value each proposed agile practice (such as pair programming, continuous integration, unit testing and refactoring) brings to the organization to achieve its business goals (Abbas et al., 2010; Ahmed and Sidky, 2009; Kurapati et al., 2012). Analysis of previous research shows that a comprehensive SLR covering agile methods tailoring doesn't exist yet. This study aims to systematically review the existing literature on agile methods tailoring, summarize

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the research and technical aspects, and identify trends and gaps in the existing research.

These types of challenges motivated academic researches on agile practice adoption and practices selection in order to generate a tailored software development method consistent with the organization's values, culture, reality, needs and strategies (Abbas et al., 2010; Ahmed and Sidky, 2009; Esfahani et al., 2011; Kurapati et al., 2012; Madi et al., 2011; Qumer and Henderson-Sellers, 2008). This way the organization will only spend effort adopting the practices that aggregate value to its strategy and help to achieve its business goals (Abbas et al., 2010; Ahmed and Sidky, 2009; Esfahani et al., 2011).

The problem of tailoring agile methods to be adopted by organizations is a known problem and researchers have been working on it but there is still no definitive answer for the theme (Abbas et al., 2010; Ahmed and Sidky, 2009; Esfahani et al., 2011; Kurapati et al., 2012; Madi et al., 2011). A secondary study such as this can help to identify problematic situations and potential solutions on the research theme. The SLR was chosen as the research method to detect trends in agile methods tailoring research and identify gaps and opportunities for further investigation (Kitchenham et al., 2009).

The objectives of this study are to summarize the existing literature on agile methods tailoring, to provide understanding how the research on agile methods tailoring is conducted (how empirical is the research, which approaches are being explored, what types of results have been generated), to identify the research community's view of agile method tailoring (which agile methods have been studied and how these approaches propose the agile practice selection), and to identify research gaps on the theme. These objectives could help organizations to understand how to fit agile practices on their context, adopt new practices with minimal disruption of the existing environment and start getting some of the benefits of agile methods.

The remainder of the paper is organized as follows. Section 2 provides the technical background and related work while Section 3 discusses the research methodology and data extraction procedures. The results of the study are presented on Section 4 and findings are discussed on Section 5. The conclusions and future work are presented on Section 6.

2. Background and related work

This section reviews two key concepts involved on this research: agile software development and software methods tailoring.

2.1. Agile software development

The Agile Manifesto (Beck et al., 2001) has consolidated values and principles from existing agile methods and approaches, and made the agile movement stronger and organized on the software development industry.

The agile values are:

- Individuals and interactions
- Working software
- Customer collaboration
- Responding to change

The agile principles are:

- Early and continuous delivery of valuable software
- Change is welcome
- Deliver frequently
- People interaction (business and developers)
- Motivated people
- Face-to-face communication
- Working software is progress
- Constant pace
- Technical excellence and good design

Table 1

15 most adopted agile practices according to Version One State of Agile 2013 Research (VersionOne, 2013).

Position	Practice	Adoption percentage (%)
1	Daily stand-up	85
2	Iteration planning	75
3	Retrospectives	74
4	Unit testing	72
5	Release planning	70
6	Burndown/Team-based estimation	69
7	Velocity	60
8	Continuous integration	58
9	Automated builds	56
10	Coding standards	55
11	Dedicated product owner	55
12	Integrated Dev/QA	50
13	Refactoring	47
14	Digital task board	45
15	Open work area	44

- Simplicity
- Self-organized teams
- Continuous improvement

Agile methods are in their essence based on values and principles defined on the Agile Manifesto (Beck et al., 2001) and composed by agile practices (Jalali and Wohlin, 2010). Agile practices should help accomplish agile principles in a method and can be grouped into management practices, software process practices and software development practices (Lee and Yong, 2013). Example of management practices are: on-site customer, daily stand-up meetings and open work area. Software process practices include simple design and collective code ownership. Pair programming and unit testing are examples of software development practices. Table 1 shows a list of most adopted practices in 2013 according to Version One State of Agile 2013 Research (VersionOne, 2013).

Handling unstable requirements, delivering working software in short time frames, with high quality and under budget are the main characteristics of agile methods compared to traditional ones (Jyothi and Rao, 2011). Being agile is to be able to rapidly adapt to change in a flexible way (Qumer and Henderson-Sellers, 2006). The capability is reflected by the attributes of flexibility, velocity, leanness, learning and response to change. According to Qumer and Henderson-Sellers (2006) agility can be defined as the ability to accommodate changes (expected or not) in a dynamic environment being simple, economic and having quality in a short iteration strategy applying previous knowledge and generating new ones on this experience. Handling unstable requirements, delivering working software in short time frames, with high quality and under budget are the main characteristics of agile methods compared to traditional ones (Jyothi and Rao, 2011).

2.2. Agile methods

The current mainstream agile methods are: Extreme Programming (XP), Scrum (Deemer et al., 2010; Schwaber and Sutherland, 2011), Kanban, Lean, Feature-Driven Development (FDD), Dynamic Systems Development Method (DSDM), Adaptive Software Development (ASD), Crystal and Rational Unified Process (RUP). Scrum is currently the most adopted one (VersionOne, 2013). Each method focuses on specific values and there is no standard on how a method should implement its agile features.

XP has been created to help small teams to develop software when requirements are vague and change frequently (Beck, 2000). It is considered a lightweight methodology and focuses on cost savings, unit tests before and during code activities, frequent full system integration, pair programming, simple design and frequent releases of working software (Beck, 2000).

Scrum is a process framework to deliver products with the highest possible value and handle complex problems or situations (Schwaber and Sutherland, 2011). Use iterative and incremental approaches to develop products using cross-functional teams (Deemer et al., 2010). Based on empiricism, Scrum focus on knowledge, experience and decision making based on what is known (Schwaber and Sutherland, 2011).

In the dawn of agile development lean manufacturing has been adapted to the software development context (Petersen, 2010). According to lean thinking the activities that do not add value to the customer are considered waste (Ikonen et al., 2010). The idea behind it, in the software development context, is to deliver features needed by the customers (Poppendieck and Poppendieck, 2003).

Kanban has been created as a lean tool to manage manufacturing operations (Ikonen et al., 2010). It has also been applied to software development and has been considered an adaptable, visual and cost focused methodology. Since it is based on lean principles Kanban tries to remove the waste of the production process. The main rules are: visualize the work flow, limit work in progress and measure the time to finish the tasks (Ikonen et al., 2010).

Feature-Driven Development (FDD) is an agile and adaptive method proposed to affect only the design and building phases of the software development life cycle (Abrahamsson et al., 2002). Has a great concern about quality of the process and frequent deliveries along the time. It is composed of five main sequential processes: develop an overall model, build a features list, plan by feature, design by feature and build by feature. FDD claims to be fit for critical systems development unlike other agile methods (Abrahamsson et al., 2002).

DSDM is an open-framework for rapid application development (RAD) and it assumes that the project flow should be based on the amount of resources and time to develop features should be predefined and the final number of features to be released should be flexible according to productivity (Abrahamsson et al., 2002).

ASD has been proposed in the year 2000 to solve issues on large and complex systems development (Abrahamsson et al., 2002). The method tries to provide guidance to avoid projects failure but at the same time not excessive guidance that could prevent creativity or cause the process to become slow and not flexible. Strong concepts related to ASD are iterative development, incremental approach for features and prototyping (Abrahamsson et al., 2002).

Crystal is a family of methodologies to allow the organization to select one of the methodologies as the appropriate for a project (Abrahamsson et al., 2002). This idea is based on contingency factors method tailoring (see Section 3.5). The methodologies are selected based on project size and how critical the project is. Each of the methodologies on the family has a color code to indicate how heavy they are (Abrahamsson et al., 2002). Crystal methodologies also use time-boxed iterations and focus on communication and collaboration.

Rational Unified Process (RUP) is an iterative process for object-oriented system development and has been created to complement UML (Abrahamsson et al., 2002). It strongly uses UML components such as use cases. The life cycle in RUP is composed of four phases: inception, elaboration, construction and transition (Abrahamsson et al., 2002). Regarding adoption it can be used in a whole or in part, depending on the organization's goals.

2.3. Software method tailoring

Software development methods have been defined with the assumption that they can be applied to any type of project and application development (Lycett et al., 2008). After failing in numerous projects the software development community has started to research and discuss software method tailoring in order to make the method more adherent to the development context (Fitzgerald et al., 2000).

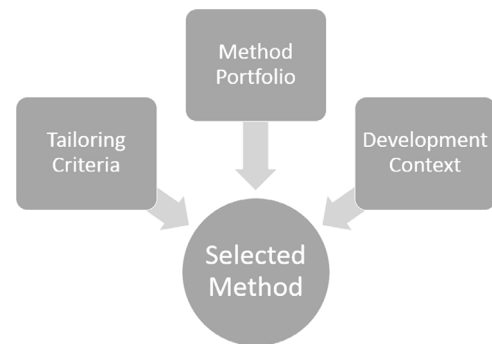


Fig. 1. Contingency factors process for method selection.

Software method tailoring has been discussed for a long time and continue to be a current need (Conboy and Fitzgerald, 2010). No matter the selected method or process, tailoring is normally needed for context adequacy in either organization and project levels (Kalus and Kuhrmann, 2013). Tailoring can be supported and stimulated by the adopted software method or executed based on the organization needs (Fitzgerald et al., 2000).

Tailoring in software process context can be defined as the adaptation of the method to the aspects, culture, objectives, environment and reality of the organization adopting it. The options to execute software method tailoring have been studied and can be mainly classified as Conboy and Fitzgerald (2010); Fitzgerald et al. (2000): contingency factors and method engineering.

2.3.1. Contingency factors

Contingency factors approach handles the tailoring of software development methods by choosing multiple methods to be available to the organization and doing the method selection based on development context features such as uncertainty level, impact and structure (Fitzgerald et al., 2000; 2003). The assumption is that there is no software development method good enough for the cases an organization can face when developing software (Conboy and Fitzgerald, 2010).

The correct method should be selected according to the development context from a portfolio of methods predefined by the organization (Fitzgerald et al., 2003) and the criteria defined by the organization as represented on Fig. 1. The main challenge for contingency factors adoption is that the team members would have to understand and be capable of execute a range of methods according to the contingencies needed for the context (Fitzgerald et al., 2003; Lycett et al., 2008).

2.3.2. Method engineering

The method engineering theory is based on meta-method processes and proposes the creation of a new method to be applied on specific contexts based on existing method fragments (Conboy and Fitzgerald, 2010). Method engineering concentrates on all activities related to software methods the same way software engineering concentrates on all activities related to software creation (Brinkkemper, 1996). It assumes the creation of organization or project specific methods and not the acquisition of an existing methods from the community or from a vendor (Henderson-Sellers and Ralyté, 2010). The idea is to create more efficient methods to respond to the challenges of real projects and their context (Brinkkemper et al., 1998).

The in-house creation of the method makes the team and the organization the owners of the method and of the fragments. This brings flexibility to the owners but also introduce challenges such as how to control the fragments or how to assemble the method for the context specific situations (Henderson-Sellers and Ralyté, 2010).

Method engineering requires a fragment repository to be operational and a method engineer to configure the method

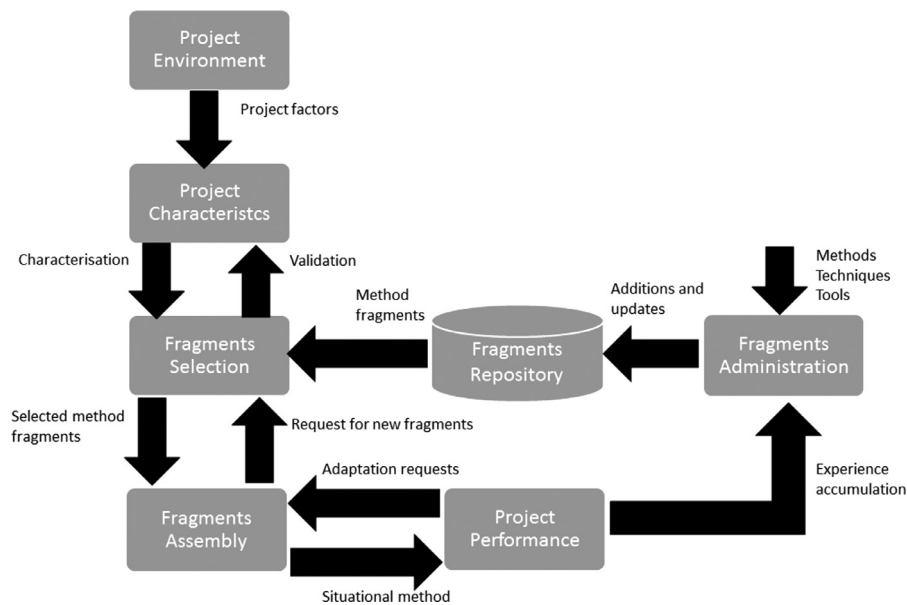


Fig. 2. Method engineering process for method definition (Brinkkemper, 1996).

Table 2

Development method tailoring criteria according to Kalus and Kuhrmann (Kalus and Kuhrmann, 2013).

Type	Examples
Team	Size, distribution, turnover, previous cooperation, good cooperation, domain knowledge, tool knowledge, technology knowledge and process knowledge.
Internal environment	Prototyping, Clear project proposal, management availability, management support, project budget, project duration, project type, project role, sub contractors, financial controlling, measurement, technical support, programming language, COTS products, operating system, database system, tool infrastructure.
External environment	Legal aspects, number of Stakeholders, stakeholder availability, stakeholder background, requirements stability, client process, client availability, type of contract, user availability, user background and trainings.
Objectives	Complexity, degree of innovation, legacy system, legacy system documentation, domain, conceptual solution, technical solution, safety & security, hardware development, neighboring systems, user interface and system integration test.

according to the proposed scenario or situation (Fitzgerald et al., 2000). Fig. 2 shows an example of a method engineering process based on Brinkkemper (Brinkkemper, 1996). Challenges for method engineering are tools for method construction, quality evaluation models for methods, knowledge on how to handle certain situations and the definition of which fragments would work best in those situations (Henderson-Sellers and Ralyté, 2010).

2.3.3. Criteria for tailoring

Another important aspect of software method tailoring is the criteria for tailoring. Kalus and Kuhrmann's work (Kalus and Kuhrmann, 2013) provided a view on the main method tailoring criteria found on the literature independently of the software method tailoring type. The criteria groups and details proposed on their work are summarized on Table 2.

The team tailoring criteria group represents aspects related to the development team and how the team is composed and interact. It

includes the team size, how the team is geographically distributed, how much of the application domain the team knows, how well they know the technologies involved on the projects and how familiar the team is with the process (Kalus and Kuhrmann, 2013).

The organizational operation and processes aspects are described on the internal environment criteria group. Management support and availability, project budgeting process, and project types are examples of important aspects for this criteria group (Kalus and Kuhrmann, 2013).

The external environment criteria group reflects the aspects outside of the organization such as laws, customer availability, type of contracts signed and requirements stability (Kalus and Kuhrmann, 2013).

The objectives criteria group represents the whole technological environment of the organization as well as business goals to be achieved. This is wide criteria group handling technical aspects such as legacy systems, user interface and system integration but also business related aspects such as degree of innovation and safety and security (Kalus and Kuhrmann, 2013).

These criteria and aspects can be used to select which method or which practices should be adopted by the organizations. The purpose of this type of analysis is to try to find the best method or set of practices for the software development context.

2.4. Related work

The problem of selecting agile practices to be adopted by organizations is a known problem and researches have been working on it but there is still no definitive answer for the theme (Abbas et al., 2010; Ahmed and Sidky, 2009; Esfahani et al., 2011; Kurapati et al., 2012; Madi et al., 2011). Not all organizations are ready to implement agile in a full manner regarding to both on cultural and technical aspects (Qumer and Henderson-Sellers, 2008). Besides these changes demand a considerable financial investment and also have a big impact on the daily tasks performed by the teams. The agile adoption could be partial and also keep other practices the organization currently adopts in place and combined with the new set of practices to make the organization more successful.

Agile methods are not always used in their full extent in an organization because of specific needs and constraints. The agile practice selection allows organizations to best achieve their goals (Kurapati et al., 2012). There is no final academic solution on how the

practice selection should happen. Organizations can adopt agile in multiple ways, depending on their business goals, culture and resources (Abbas et al., 2010). The agile practice adoption can also be unique per organization taking into account the areas of the development process the organization wants to improve. Normally organizations need help choosing the right combination of practices for their environment. The association/correspondence between agile values and the organization culture defines which set of practices could be followed by this organization (Madi et al., 2011).

The research work on agile methods tailoring can be divided on the following types: most used practices, quality, business goals, maturity model, agile values and project types.

A systematic review of the research literature on use of agile in global software engineering presented a list of agile practices used or referenced by the analyzed papers (Jalali and Wohlin, 2010). A survey on agile practice adoption proposed to understand which practices have been used by the software development industry (Kurapati et al., 2012). This survey analyzed practices adoption at the project and organization levels and also defined the association between practices. Factor analysis was applied to group 58 agile practices and found 15 factors and checked the correlation between them finding application of quality assurance and iterative and incremental practices had a high success rate (Abbas et al., 2010). An empirical study identified common adopted practices, listed practices that normally are used together and correlated the customer satisfaction after adopting the practices (Manyam and Kurapati, 2012). This work also tried to identify adaptation by organizations on the adopted practices.

A work focused on the quality aspects of the agile practices showed the main agile practices associated with high quality of the software product (de Azevedo Santos et al., 2011).

SAAP (Strategic Analysis for Agile Practices) framework (Esfahani et al., 2011) was proposed with the intent to associate business goals of the organization to the selection process for agile practice adoption. SAAP extended Situation Method Engineering and used concepts from Balanced Score Card (BSC). This work linked the selection of agile practices to the organization's business goals.

Sidky Agile Measurement Index (SAMI) (Ahmed and Sidky, 2009) showed the adoption of agile practices based on an agile maturity model. SAMI is a 5-step road map to guide teams adopting based in five values considered essential to agility: enhancing communication and collaboration (level 1), delivering software early and continuously (level 2), developing high quality, working software in an efficient and integrated manner (level 3), respond to change through multiple levels of feedback (level 4) and establishing an environment to sustain agility (level 5). SAMI is not based on any specific agile method such as XP, Scrum or Crystal, but instead, uses agile values and principles to define the path to agility.

The analysis of papers and books from Agile Manifesto signatories using content analysis (Madi et al., 2011) identified the key agile values and how frequent they were mentioned in the literature. The agile key values obtained by this work were: Flexibility, Customer-centric, Working Software, Collaboration, Simplicity, Communication, Natural, Learning, Pragmatism and Adaptability.

A methodology to select the best agile practices for projects based on the association between project's characteristics and the abilities of the agile practices has been defined (Saleh, 2013) and the key project's areas analyzed in the work were team size, iteration duration and distributed teams.

3. Research

Evidence-based software engineering (EBSE) has been a growing trend in the software engineering field since the article from Kitchenham et al. in 2004 (Kitchenham et al., 2004). The implementation of EBSE include the conversion of information needed into

Table 3
Research questions for the SLR.

ID	Question
RQ1	What are the methodological aspects of agile method tailoring research?
RQ2	How practical has agile methods tailoring research been?
RQ3	Which are the method tailoring criteria used for agile methods tailoring?

questions to be answered, find the evidence to answer the questions and appraise the evidence found in a critical way (Brereton et al., 2007). These three steps constitute a SLR. As more researchers seem to be interested on EBSE approach and in the SLR methodology, the software engineering SLR's quality has improved over the years (Brereton et al., 2007; Kitchenham et al., 2009).

SLR was chosen as the research method for this study to allow evidence gathering on the agile methods tailoring literature to provide guidelines for researchers on research trends and gaps, and to identify guidelines for agile practitioners (Kitchenham, 2004; Kitchenham et al., 2009).

The design of this SLR was based on the guidelines provided by Kitchenham et al. (Kitchenham, 2004; Kitchenham et al., 2009), Brereton et al. (Brereton et al., 2007) and Petersen et al. (Petersen et al., 2008). The steps of the review were: definition of the research questions and the research protocol, conduct search for studies, screening of papers, data extraction and data synthesis.

3.1. Research questions

The research questions identify the scope of the SLR, guide the research protocol construction and the most critical elements of the SLR (Brereton et al., 2007). The research questions were identified according to the focus of this study and listed on Table 3. The main goal of the research questions is to provide an understanding of the context of agile methods tailoring research in the literature.

Research question 1 (RQ1) aims to identify the methodological aspects of agile methods tailoring research papers. This research question considers three facets (empirical research, research validation and research type) in order to define the methodological aspects of the papers. The results of this research question can help to understand how agile methods tailoring research has been conducted and also provide guidance to improve its scientific robustness.

The intention of research question 2 (RQ2) is to identify how close to practice is agile methods tailoring research. In order to obtain the desired answer, the analysis takes into account the following facets: agile methods, method tailoring approaches and research results. This provides a broader view on how practical the research has been and how applicable to real world organizations the research can be.

The proposition of research question 3 (RQ3) is to characterize how papers researching agile methods tailoring proposed the criteria used for method tailoring. In this question's case, that should point out if papers take into account, for instance, project budget or team size as parameters to tailor the agile method. It considers three facets: tailoring criteria group, method tailoring approaches and agile methods. The result of the question can be considered a trend on agile methods tailoring research and indicate where the majority of the research papers have been heading to on regards of the three facets composing this analysis (tailoring criteria group, method tailoring approaches and agile methods).

3.2. Search strategy

To conduct the search for the papers, first we defined the search terms and created the search string. The second step consisted on the definition of the digital libraries and the data retrieval settings on the

Table 4
Search strings for data retrieval.

(agile AND practice AND select) OR
(agile AND method AND tailoring) OR
(scrum AND practice AND adoption) OR
(scrum AND tailoring) OR
(scrum AND practice AND select) OR
(xp AND practice AND adoption) OR
(xp AND tailoring) OR
(xp AND practice AND select) OR
(kanban AND practice AND adoption) OR
(kanban AND tailoring) OR
(kanban AND practice AND select) OR
(lean AND practice AND adoption) OR
(lean AND tailoring) OR
(lean AND practice AND select) OR
(fdd AND practice AND adoption) OR
(fdd AND tailoring) OR
(fdd AND practice AND select) OR
("feature driven development" AND practice AND adoption) OR
("feature driven development" AND tailoring) OR
("feature driven development" AND practice AND select)

selected digital libraries. The third step was the definition and application of the inclusion and exclusion criteria on the search results.

3.2.1. Search terms

The search terms were defined based on agile method tailoring, variations of popular agile methods (VersionOne, 2013) and also "practice adoption" which is a term used in the literature as a synonym for "agile method tailoring" (Abbas et al., 2010; Esfahani et al., 2011; Jyothi and Rao, 2011; Kurapati et al., 2012). Table 4 summarized the search strings.

3.2.2. Data retrieval

The digital databases chosen for data retrieval were:

- ACM Digital Library
- IEEE Xplore
- Science Direct
- Springer
- Wiley Online Library

The data retrieval process consisted of the search execution using the search strings on each of the digital libraries and the retrieval of the papers and their meta-data into a reference management tool named Zotero¹.

3.2.3. Selection criteria

We defined the inclusion and exclusion criteria summarized on Table 5 in order to filter irrelevant papers out.

3.3. Screening of papers

The search on the digital libraries (stage 1) identified a total of 783 papers. The screening of the papers selected the appropriate studies based on the research theme using the selection criteria. The selection criteria was composed by five filter stages as shown of Fig. 3. Stage 2 removed the duplicated references to papers found on multiple digital libraries and resulted in 699 papers. At stage 3 a filter by publication date (from 2002 to May 2014), type of article (academic papers) and type of periodical (journals or conferences related to computer science, software development and software engineering) according to the selection criteria was applied and removed 24.89% of the papers, reducing the number of papers to be analyzed to 525. The filter by article title was manually applied on stage 4, the first

Table 5
Inclusion and exclusion criteria for papers selection.

Inclusion criteria	Exclusion criteria
Article date: publications from 2002 to May 2014. The start year is one year after the release of the Agile Manifesto (Beck et al., 2001) and publications after that should start to reference the Agile Manifesto, agile methods and tailoring approaches.	Books, thesis, editorials, prefaces, article summaries, interviews, news, reviews, correspondence, discussions, comments, reader's letters and summaries of tutorials, workshops, panels, and poster session.
Academic papers published on journals or conferences related to computer science, software development and software engineering.	Studies not covering agile methods or practices adoption or tailoring on software development.
Papers should describe agile methods tailoring, agile methods assessment, agile practice selection or agile adoption approaches.	References to studies downloaded from the digital libraries missing the abstract or studies missing the abstract.
	Duplicate papers found on the digital libraries.

author has analyzed all articles by title to select the ones appropriate according to the inclusion criteria and a total of 336 papers were selected for further analysis. At stage 5 the filter by article abstract was applied by the first author to provide a more detailed analysis on the papers and removed studies not covering agile methods tailoring theme reducing the total number of papers to 56. At the end of the screening process 7.15% of the papers identified by the search on the digital libraries have been selected to be fully read and go through data extraction and classification procedures.

3.4. Data extraction

The data extraction and classification was based on Petersen et al. (Petersen et al., 2008) and adjusted to the research questions and relevant data for this study. The selected 56 papers were full read and both text and meta-data were used to classify the papers according to the defined categories.

3.5. Classification

The classification categories utilized on this paper were: research type, empirical study type, research questions type, research results type, research validation type, supported agile methods, method tailoring approach and method tailoring criteria. During the data extraction, studies were classified into more than one item on the following categories: supported agile methods and method tailoring criteria.

Research types: the research types were based on Wieringa et al. (Wieringa et al., 2006) and described on Table 6: validation research, evaluation research, solution proposal, philosophical papers, opinion papers and experience papers.

Empirical study types: the following types of empirical research were considered (Tonella et al., 2007): experiment, case study, observational study, experience report and systematic review. Table 7 provides a brief description of the types.

Research questions types: method or means of development; method for analysis or evaluation; design, evaluation, or analysis of a particular instance; generalization or characterization and feasibility study or exploration. The types were based on Shaw (Shaw, 2003) and summarized on Table 8.

Research results types: the research result types used on this paper were defined by Shaw (Shaw, 2003): procedure or technique; qualitative or descriptive model; empirical model; analytic model; tool or notation; specific solution, prototype, answer, or judgment; report. Table 9 describes the types.

¹ Available at <https://www.zotero.org/download/>.

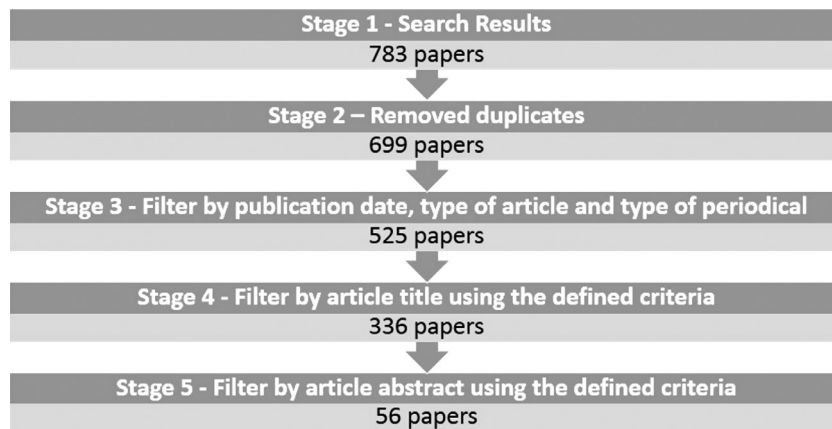


Fig. 3. Paper screening process from SLR search execution to final selection.

Table 6

Research types according to Wieringa et al. (Wieringa et al., 2006).

Type	Description
Validation research	Investigates proposed techniques not yet implemented using a strong research method. The technique may have been proposed on a different paper and by any author.
Evaluation research	Involves the practical investigation or implementation of a technique. Normally results in a new knowledge claim around the studied phenomena and that claim should be evaluated on the research.
Solution proposal	Presents a technique, normally a novel or an improvement of an existing technique, and justifies its relevance. It is not necessary to completely implement the technique but a proof of concept could be presented.
Philosophical papers	Conceptual frameworks or a new point of view on existing techniques or approaches.
Opinion papers	Personal opinion of the authors about a technique highlighting the advantages and disadvantages of it. Not supported by related work and research methods.
Experience papers	Relates author's experience on implementation and practical use of techniques. Should contain the lessons learned from the experience.

Table 7

Empirical study types according to Tonella et al. (Tonella et al., 2007).

Type	Description
Experiment	Controlled study to observe the outcomes and the factors involved on it. Requires observation of multiple cases.
Observational study	Gathers information to connect factors and effect variables. It is not controlled and requires observation of multiple cases. Normally is applied as a survey to collect data to be observed.
Experience report	Analysis of one case and there is no controlled context. The goal is to describe the success of the proposed/adopted technique and setup, data collection and analysis are not discussed in details.
Case study	Analysis of one case and setup, data collection and analysis are discussed in details.
Systematic review	Selects cases of the topic of interest studied in the past for evaluation and interpretation.

Table 8

Research questions types according to Shaw (Shaw, 2003).

Type	Description
Method or means of development	How can we do/create/modify/evolve (or automate doing) X? What is an improved way to do/create/modify/evolve X?
Method for analysis or evaluation	How can I evaluate the quality/correctness of X? How do I choose between X and Y?
Design, evaluation, or analysis of a particular instance	How good is Y? What is property X of artifact/method Y? What is a (better) design, implementation, maintenance, or adaptation for application X? How does X compare to Y? What is the current state of X / practice of Y?
Generalization or characterization	Given X, what will Y (necessarily) be? What, exactly, do we mean by X? What are its important characteristics? What is a good formal/empirical model for X? What are the varieties of X, how are they related?
Feasibility study or exploration	Does X even exist, and if so what is it like? Is it possible to accomplish X at all?

methods (VersionOne, 2013): Scrum, Extreme Programming (XP), Lean, Kanban and Feature-Driven Development (FDD). During the paper analysis we also found references to other agile methods and included them on our list: Crystal, RUP, Adaptive Software Development (ASD) and Dynamic Systems Development Method (DSDM).

Method tailoring approaches: contingency factors and method engineering. The contingency factors approach assumes that the development team has defined a portfolio of development methods to be available to be chosen according to project context and features needed (Conboy and Fitzgerald, 2010; Fitzgerald et al., 2000). Method engineering is a meta-method process based on method tuning to specific project needs or situations. The method is re-created according to the defined parameters and it is composed of method fragments or chunks previously proven to work on the currently faced situation (Conboy and Fitzgerald, 2010; Fitzgerald et al., 2000).

Method tailoring criteria: the criteria used for the method tailoring have been defined on Kalus and Kuhrmann (Kalus and Kuhrmann, 2013) and grouped in 4 categories: team, internal environment, external environment and objectives. The criteria categories and examples of items inside each category can be seen on Table 2. During the data gathering phase the authors noticed that there were criteria used on the articles but not listed on Kalus and Kuhrmann work (Kalus and Kuhrmann, 2013). Based on that, the article analysis was also conducted using an extended criteria category list described on Table 11 with two new criteria types defined by the authors (maturity levels and previous projects) and some new examples of criteria for existing types.

Research validation types: analysis, evaluation, experience, example, persuasion and blatant assertion. The types were based on Shaw (Shaw, 2003). Examples of each type are shown on Table 10.

Supported agile methods: each of the tailoring techniques could be applicable to specific agile methods. We listed popular agile

Table 9
Research results types according to Shaw (Shaw, 2003).

Type	Examples
Procedure or technique	New or improved way to do a task, such as design, implementation, maintenance, measurement, evaluation, selection from alternatives; includes techniques for implementation, representation, management, and analysis; a technique should be operational, not advice or guidelines, but a procedure
Qualitative or descriptive model	Structure or taxonomy for a problem area; architectural style, framework, or design pattern; non-formal domain analysis, well-grounded checklists, well-argued informal generalizations, guidance for integrating other results, well-organized interesting observations
Empirical model	Empirical predictive model based on observed data
Analytic model	Structural model that permits formal analysis or automatic manipulation
Tool or notation	Implemented tool that embodies a technique; formal language to support a technique or model (should have a calculus, semantics, or other basis for computing or doing inference)
Specific solution, prototype, answer, or judgment	Solution to application problem that shows application of SE principles, may be design, prototype, or full implementation; careful analysis of a system or its development, result of a specific analysis, evaluation, or comparison
Report	Interesting observations, rules of thumb, but not sufficiently general or systematic to rise to the level of a descriptive model

Table 10
Research validation types according to Shaw (Shaw, 2003).

Type	Examples
Analysis	Applies rigorous analysis based on data gathered on models and/or experiments
Evaluation	Validation is based on a previous defined criteria or done by subject matter experts
Experience	Real world results generated by someone other than the author can provide evidence of success
Example	Focuses on an example to define how the real world problem can be resolved
Persuasion	It is more about the idea than the results and normally has a lot of passion associated to it
Blatant assertion	No results have been evaluated

4. Results

This section presents the descriptive results of the SLR.

When extracting data from the papers and classifying it, the year of publication was an useful information to understand how active is the research on the agile methods tailoring theme. Fig. 4 shows the number of articles published on the theme per year. There was no paper publication on the years following the release of the Agile Manifesto (Beck et al., 2001) between 2002 and 2003 and only one paper was published in 2004. This seems to indicate that the researchers focus on agile methods tailoring is recent and suggests that agile methods tailoring has become a more active research theme only after agile methods have been adopted and tested in practice and the challenges on adopting it have been studied. Year 2013 was the year with the majority of the published papers (11 papers).

The geographical distribution of agile methods tailoring papers over the years is presented on Fig. 5. It shows that there is no concentration of papers in a country on a given year and that

Table 11
Development method tailoring criteria based on Kalus and Kuhrmann (Kalus and Kuhrmann, 2013) and extended by the authors.

Type	Examples
Team	Examples from Kalus and Kuhrmann (2013): Size, distribution, turnover, previous cooperation, good cooperation, domain knowledge, tool knowledge, technology knowledge and process knowledge.
Internal environment	Examples from Kalus and Kuhrmann (2013): Prototyping, Clear project proposal, management availability, management support, project budget, project duration, project type, project role, sub contractors, financial controlling, measurement, technical support, programming language, COTS products, operating system, database system, tool infrastructure. Examples defined by the authors: organization size, communication aspects, culture.
External environment	Examples from Kalus and Kuhrmann (2013): Legal aspects, number of Stakeholders, stakeholder availability, stakeholder background, requirements stability, client process, client availability, type of contract, user availability, user background and trainings.
Objectives	Examples from Kalus and Kuhrmann (2013): Complexity, degree of innovation, legacy system, legacy system documentation, domain, conceptual solution, technical solution, safety & security, hardware development, neighboring systems, user Interface and system integration test. Example defined by the authors: business goals.
Maturity levels	Example defined by the authors: maturity levels to define which agile practices should be adopted.
Previous knowledge	Example defined by the authors: work done on previous projects.

researchers around the world have been working on agile method tailoring theme². The main countries generating papers about agile methods tailoring over the years were USA, UK, Finland and Ireland but there is no specific country dominating the research scenario on the theme among the 23 countries listed.

5. Discussion

This section describes and discusses the findings from the data extraction and classification activities. The findings are presented in a graphical view and are organized by research question.

5.1. RQ1 - What are the methodological aspects of agile method tailoring research?

The intent of RQ1 is to show the methodological aspects used on agile methods tailoring research. Empirical research was used on 66.1% of the analyzed papers and the most used types of empirical research were case study (35.7%) and observational study (25.0%). The results summarized on Table 12 show 19 papers (33.9%) classified as “None” meaning they didn’t use any type of empirical research procedure.

Regarding to the research type used by the analyzed papers (see Table 13), solution proposal is the main research type adopted (42.8%) and experience papers is the second one (26.8%). This shows that the majority of the papers are either proposing new approaches for agile methods tailoring or reporting approaches already used in practice but not necessarily created based on scientific methods.

² The country of the paper was defined by the first affiliation of the first author.

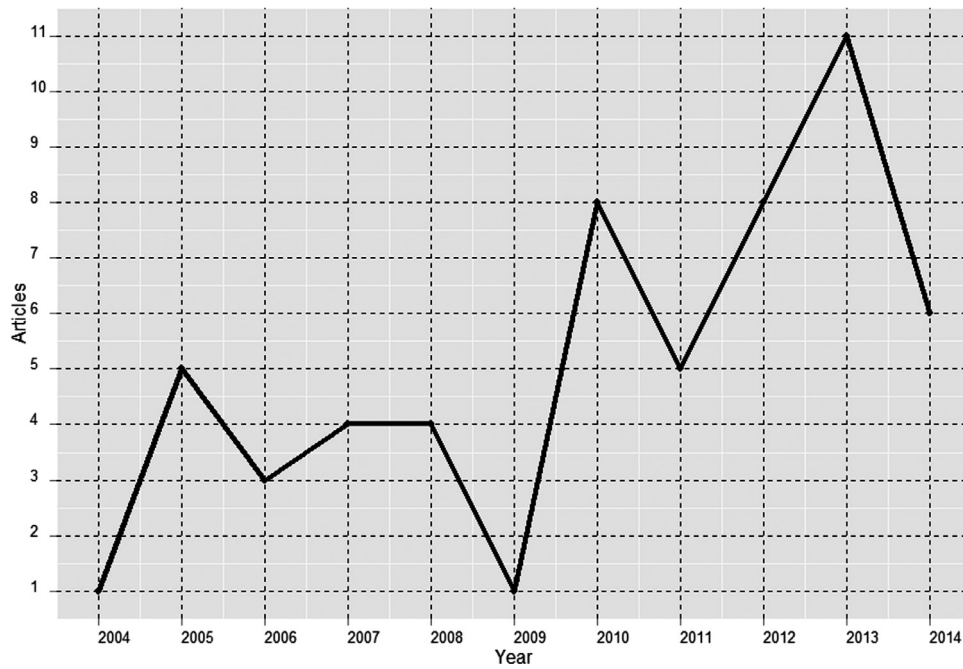


Fig. 4. Number of papers on agile methods tailoring per year.

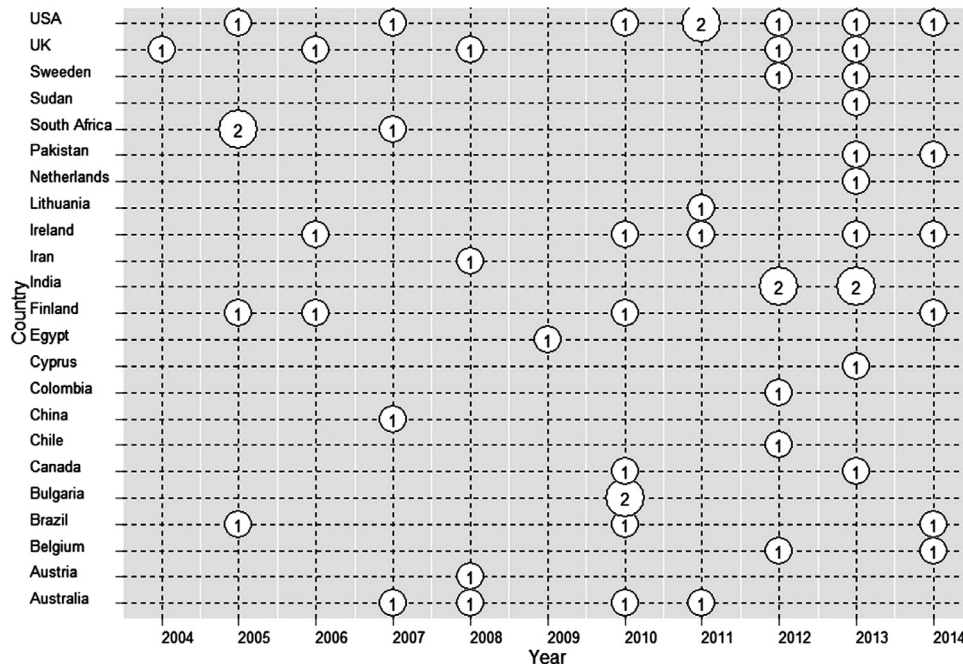


Fig. 5. Geographical distribution of agile methods tailoring papers over the years.

Table 12

Empirical research type usage by the analyzed papers.

Empirical research	Number of articles	Percentage (%)
Case study	20	35.7
None	19	33.9
Observational study	14	25.0
Systematic review	2	3.6
Experiment	1	1.8

Table 13

Research type usage by the analyzed papers.

Research type	Number of articles	Percentage (%)
Solution proposal	24	42.8
Experience papers	15	26.8
Evaluation research	8	14.3
Philosophical papers	8	14.3
Validation research	1	1.8

The research validation facet shows how strong is the validation of the research executed on the analyzed papers and the results are summarized on Table 14. Evaluation is the main type of research validation on the analyzed papers (57.1%) followed by example and

persuasion (both with 16.1%). Two of the papers (3.6%) didn't even try to validate their work and were classified as blatant assertion regarding to the research validation facet. The research validation on the analyzed studies is mainly based on example and persuasion and

Table 14
Research validation usage by analyzed papers.

Research validation	Number of articles	Percentage (%)
Evaluation	32	57.1
Example	9	16.1
Persuasion	9	16.1
Experience	3	5.4
Blatant assertion	2	3.6
Analysis	1	1.8

Table 15
Agile methods usage by analyzed papers.

Agile method	Number of articles	Percentage (%)
Non-specific	30	53.6
Scrum	19	33.9
XP	19	33.9
DSDM	6	10.7
Crystal	4	7.1
ASD	3	5.4
FDD	3	5.4
RUP	2	3.6
Kanban	1	1.8
Lean	1	1.8

the research types are solution proposal and philosophical papers. An aspect that is not present on the majority of the philosophical papers is the theoretical validation of the ideas or conceptual framework proposed by a mapping study or a SLR. 80.4% of the analyzed studies presented types of validation considered acceptable for software engineering papers (evaluation, example, experience and analysis) (Shaw, 2003). Two papers did not explicitly justify or tried to validate even theoretically their proposed approaches and were classified as blatant assertion for this facet (Krasteva and Ilieva, 2010; Mnkandla, 2005).

Methodological aspects are important factors when considering writing papers to lead to a structured article, provide consistent results and contributions to the existing knowledge on software engineering (Shaw, 2003) and for the analyzed studies it can be summarized on Fig. 6. The majority of the analyzed papers (66.1%) used empirical research to get to the reported results. That is a good sign of serious research on the agile methods tailoring theme however the number of studies using empirical should be increased to improve the quality of the publications and also the change of acceptance on important conferences and journals. The number of papers not based on any empirical research procedure is considerable on the list of analyzed papers revealing that even though the majority of these papers propose approaches for agile methods tailoring they do not provide empirical evidence to support their claims. Case study was the most utilized empirical research technique and demonstrated that those researchers were trying to prove that the proposed approaches could be used on real case and with the feedback obtained by the case studies the approaches could be improved. Experiments are rare among the analyzed papers and the cause could be the lack of wide real adoption of the agile method tailoring technique, since experiments should have to involve multiple cases in a controlled setup (Tonella et al., 2007).

5.2. RQ2 - How practical has agile methods tailoring research been?

Research question 2 (RQ2) proposes a discussion on how practical is agile methods tailoring research. Fig. 7 shows the correlation between data from agile method, research results and method tailoring approaches facets. More than half of the analyzed papers (53.6%) are non-specific to any agile method (see Table 15). This data indicates that the majority of the research on agile methods tailoring could be applied to organizations independently of the agile method chosen. The rest of the papers use agile methods tailoring approaches

Table 16
Method tailoring approaches usage by analyzed papers.

Research type	Number of articles	Percentage (%)
Method engineering	39	69.7
Unclear	12	21.4
Contingency factors	5	8.9

mainly focused on Scrum (33.9%) or XP (33.9%).³ Real world organizations have been adopting mainly Scrum and XP (VersionOne, 2013) as their agile methods and this shows research connection with practical agile adoption. The analyzed agile method tailoring procedures or techniques that were non-specific worked based on a set of agile practices no matter the agile method chosen by the organization and on a model to select the practices to be adopted. There were also approaches combining agile methods to try to get the best of different agile methods (Ayed et al., 2012; El-Said et al., 2009; Meng et al., 2007).

Based on the software process tailoring articles (Conboy and Fitzgerald, 2010; Fitzgerald et al., 2000) we found two types of approaches (method engineering and contingency factors) to be used on the data classification activity but when extracting the data from the selected papers there were papers with no clear definition of method tailoring approach being used. Those papers were classified as “Unclear”. In this analysis an unclear approach meant: there was no explicit reference to the literature describing the tailoring approach and the method tailoring approach was neither method engineering or contingency factors. In these articles the tailoring seems to happen based on arbitrary criteria defined by the researchers. There was no explanation on why and how the practices were selected during the tailoring process and no literature references to indicate existing approaches being used for tailoring. According to the results (see Table 16) the method engineering approach is used in 39 (69.7%) of the selected papers, 5 of the selected papers (8.9%) used the contingency factors approach and the other 12 papers (21.4%) used an unclear approach for method tailoring.

Method engineering was utilized on the majority of the analyzed papers as the method tailoring approach. During data extraction the approach was identified by the usage of method approach definitions cited directly on the studies or detected by the authors during paper analysis. Among the studies using method engineering approach there were two based on situational method engineering (SME) (Ayed et al., 2012; Qumer and Henderson-Sellers, 2008). SME proposes the creation of software development methods taking into consideration the context of project and/or organization being analyzed. The percentage of papers with unclear method tailoring approach definition (21.4%) illustrates the broad range of methods tailored to solve specific problems in the organization and not design to be applied on different organizations (Bass, 2013; Bhalarao and Ingle, 2012; Coelho et al., 2005; Hayata and Han, 2011; Jonsson et al., 2012; McHugh et al., 2014; Rodriguez et al., 2014; Rohunen et al., 2010; Sultana et al., 2014) or touching the agile method tailoring theme in an explanatory or conceptual way (Papatheocharous and Andreou, 2013; de Souza Mariz et al., 2010; Williams, 2010).

The research results facet showed a concentration of papers generating reports (41.1%) and procedures (32.1%) as shown on Table 17. These results point into the direction of a practical approach for the research (resulting in reports of usage/experiences and procedures to be adopted) and show that research is connected the practice on the agile methods tailoring field.

The majority of papers use method engineering as the method tailoring approach and are concentrated between Scrum, XP and non-specific method regarding to the agile method utilized and between

³ The classification of papers based on agile methods allowed the papers to be classified as using multiple agile methods unless it used the non-specific option.

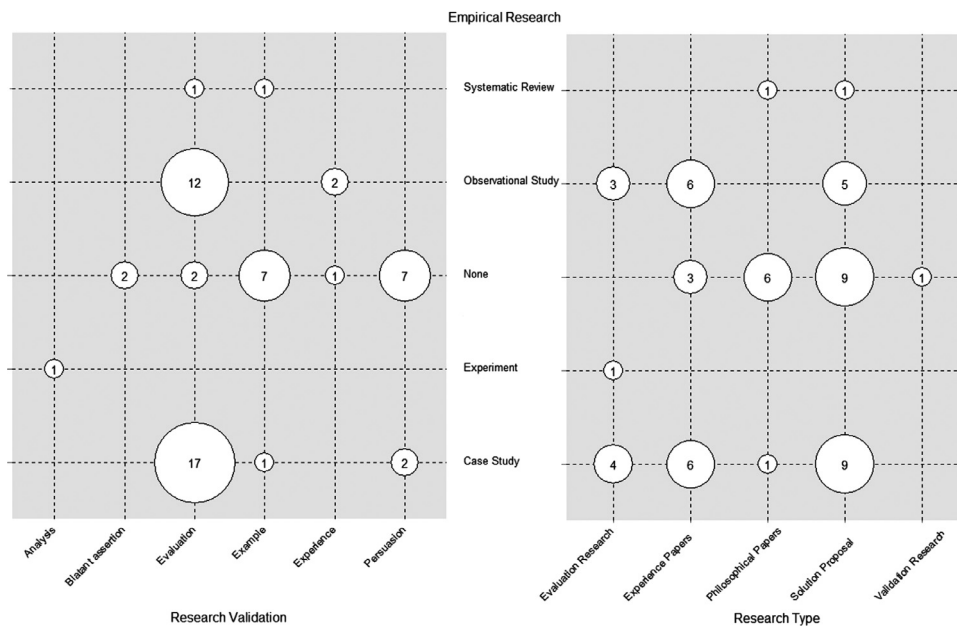


Fig. 6. Map of methodological aspects based on empirical research, research validation and research type. Empirical research types are represented by the Y axis, research validation types by the left side of the X axis and research types by the right side of the X axis.

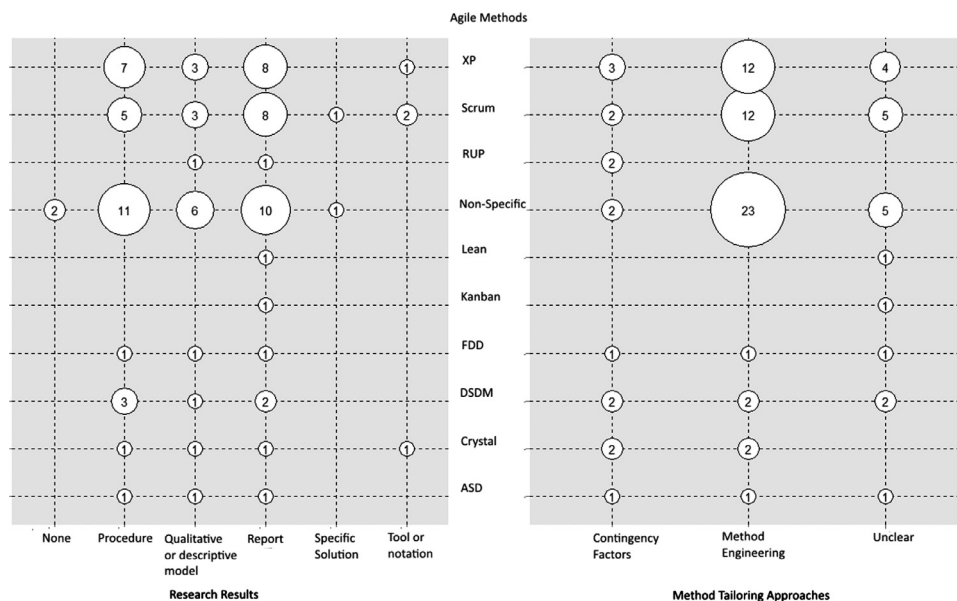


Fig. 7. Map of research based on agile methods, research results and method tailoring approaches. Agile methods are represented by the Y axis, research results by the left side of the X axis and method tailoring approaches by the right side of the X axis.

reports, procedures and qualitative or descriptive models regarding to the research results.

5.3. RQ3 - Which are the method tailoring criteria used for agile methods tailoring?

When analyzing the criteria used for agile method tailoring based on criteria groups⁴ defined by software method tailoring (Kalus and Kuhrmann, 2013) the results are disappointing showing that 42.9% of the analyzed papers cannot be classified using those criteria groups (see Analysis 1 column on Table 18). On this scenario the Internal Environment (51.8%) and the Team (33.9%) criteria group were the

most utilized criteria for method tailoring. Regarding to the criteria details inside the groups, the Project Type (46.4%) and Complexity (26.8%) were the most utilized criteria as summarized on Table 19. An alternative scenario for method tailoring criteria has been built integrating the criteria found on the literature (Kalus and Kuhrmann, 2013) with the criteria (groups and detailed items) detected by the authors during paper analysis. In this new scenario, 98.2% of the analyzed papers could be classified using the criteria groups (see Analysis 2 column on Table 18). Checking the criteria details on Table 20 the Project Type (46.4%) and Business Goals (42.9%) criteria were the most adopted ones.

For Analysis 2 we proposed two new criteria groups: Previous Knowledge and Maturity Level. The Previous Knowledge group is the criteria based on work done on previous projects and experiences of the organization with tailoring. The Maturity Level group is

⁴ Method tailoring criteria are organized in groups and detailed criteria according to Kalus and Kuhrmann (Kalus and Kuhrmann, 2013).

Table 17
Research results usage by the analyzed papers.

Research result	Number of articles	Percentage (%)
Report	23	41.1
Procedure or technique	18	32.1
Qualitative or descriptive model	9	16.1
Specific solution, prototype, answer, or judgement	2	3.6
Tool or notation	2	3.6
None	2	3.6

Table 18
Method tailoring criteria groups usage by analyzed papers.

Group	Analysis 1 (%)	Analysis 2 (%)
External environment	26.8	26.8
Internal environment	51.8	67.9
Objectives	26.8	66.1
Team	33.9	33.9
None	42.9	1.8
Previous knowledge	N/A	7.1
Maturity level	N/A	7.1

Table 19
Criteria used for agile methods tailoring detected on the analyzed papers based on Kalus and Kuhrmann (Kalus and Kuhrmann, 2013).

Criteria	Percentage of papers (%)
Project type	46.4
Complexity	26.8
Team size	25.0
Technology knowledge	23.2
User availability	16.1
Requirements stability	16.1
Team distribution	14.3
Management support	8.9
Degree of innovation	7.1
Domain knowledge	5.4
Project budget	5.4
Type of contract	3.6

Table 20
Criteria used for agile methods tailoring detected on the analyzed papers based on Kalus and Kuhrmann (Kalus and Kuhrmann, 2013) and on the authors.

Criteria	Percentage of papers (%)
Project type	46.4
Business goals	42.9
Complexity	26.8
Team size	25.0
Technology knowledge	23.2
User availability	16.1
Requirements stability	16.1
Organization size	16.1
Culture	16.1
Team distribution	14.3
Management support	8.9
Degree of innovation	7.1
Previous projects	7.1
Maturity level	7.1
Domain knowledge	5.4
Project budget	5.4
Communication	5.4
Type of contract	3.6

the criteria based on the definition and adoption of practices level based on maturity level the organizations would like to achieve when adopting agile methods. Also a few items have been added on the groups proposed by Kalus and Kuhrmann (Kalus and Kuhrmann, 2013). Business goals has been added to the Objectives criteria group and organization size, communication aspects and culture have been added to the Internal Environment criteria group. The extended criteria for Analysis 2 based on Kalus and Kuhrmann (Kalus and Kuhrmann, 2013) and containing the items added by the authors has been described on Table 11 with examples. Based on the need to provide a better classification for the analyzed articles and the in depth understanding of which criteria has been adopted for method tailoring on the analyzed papers the authors decided to use Analysis 2 (considering the proposed criteria groups and detailed items) on the results of this study. The results so far indicate that since research on agile methods tailoring is still evolving and its focus has been on practical usage (based on the results produced by the analyzed research papers) the criteria used for method tailoring is an area that is not completely developed from both the research and practical stand points and that new drivers might show up and new criteria can be created as organizations face new challenges and new scenarios for software development.

RQ3 objective is to show which are the criteria groups used on agile methods tailoring papers in order to define the tailoring strategy to be executed and select agile practices for adoption. Fig. 8 shows a concentration of papers towards internal environment and objectives groups on the tailoring criteria groups facet. This indicates that internal environment criteria such as project or management support, and objectives criteria such as business goals and organization's degree of innovation were used to define how method tailoring was implemented on the analyzed papers. Other important criteria groups are team and external environment⁵.

Regarding to the agile methods facet, the majority of the articles either use non-specific method, Scrum or XP (see Fig. 8). For Scrum and XP the most important method tailoring criteria groups were internal environment and objectives. For papers non-specific to an agile method the most relevant method tailoring criteria groups were also internal environment and objectives followed by the team criteria group.

Analyzing the method tailoring approach facet on Fig. 8 method engineering approach was the most used on the papers and the tailoring criteria groups for those papers were: internal environment, objectives, team and external environment.

5.4. Limitations

Even though the systematic process for writing SLRs is designed to avoid bias, the possibility of bias is the main limitation of these types of studies (Kitchenham, 2004). Besides the items defined by Kitchenham et al. (Kitchenham et al., 2009), we understand that the risk of bias in the study resides mainly in the paper screening and analysis process that was done by the first author. In order to address the bias threat we had a senior researcher with broad knowledge on the subject and on the SLR methodology supervising the work.

According to Kitchenham et al. (Kitchenham et al., 2009) the limitations of the SLR research method can be addressed by using a set of four quality questions:

- Are inclusion and exclusion criteria defined and appropriate?
- Is the literature search likely to have covered all relevant studies?
- Did the reviewers assess quality/validity of included studies?
- Was the basic data of the studies adequately described?

⁵ A paper could be classified as using multiple criteria groups on the tailoring criteria groups facet.

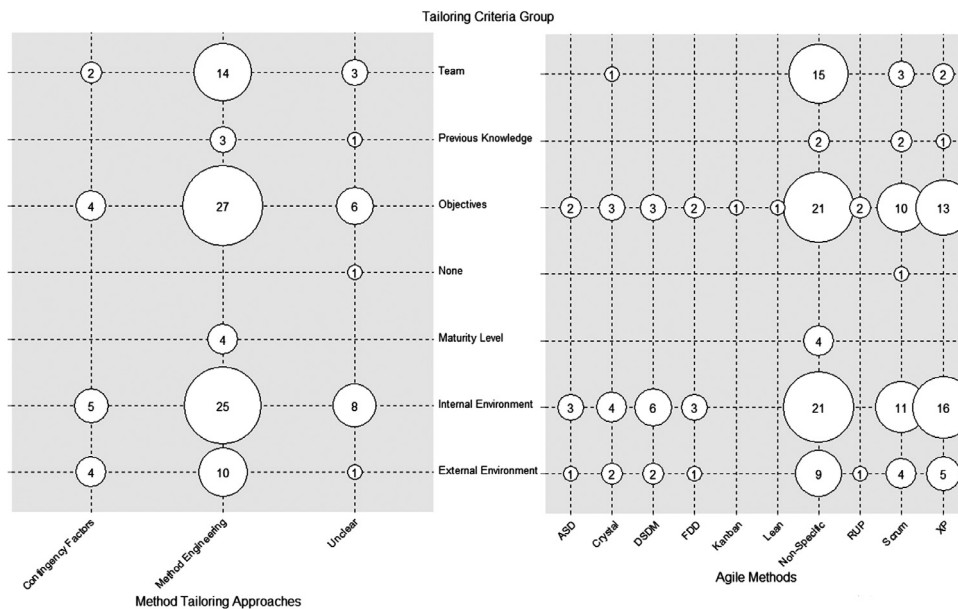


Fig. 8. Map of research based on tailoring criteria groups, agile methods and method tailoring approaches. Tailoring criteria groups are represented by the Y axis, agile methods by the left side of the X axis and method tailoring approaches by the right side of the X axis.

The inclusion and exclusion criteria have been defined on the research protocol. It was used as a tool to minimize the possibility of bias (Kitchenham, 2004) with all the exclusion and inclusion criteria being defined and revised during the execution of the phases of the SLR.

The search phase followed the search protocol to address conclusion validity and the papers have been collected from five different sources (ACM Digital Library, IEEE Xplore, Science Direct, Springer and Wiley Online Library) that would be more than the four sources mentioned by Kitchenham et al. (Kitchenham et al., 2009). The search keywords used represent different terminologies used by researchers and practitioners and the search period between 2002 and 2014 allowed the capture of relevant papers since the Agile Manifesto release however some papers may have been missed. There is no claim on collecting all relevant studies on the subject but we've collected as many studies as possible.

To address the quality of the papers we defined on our search protocol that only papers from conferences and journals would be included to try to get the best quality papers for the analysis. A detected limitation of this study is the lack of quality criteria on the paper analysis.

The data description concern was addressed during data extraction and classification stages. We have applied standard classifications defined on existing literature to allow reproducibility of the study. A perceived limitation was that a portion of the analyzed papers did not provide enough explicit details and some data has been derived from the text according to the first author's analysis.

In summary, we can claim that the patterns identified by the findings of this study will be the same as those identified by similar studies even though there might be a slightly variation on the numbers presented.

6. Conclusion and future work

The current state of the agile methods tailoring literature was summarized on this study. This SLR identified 783 studies on the search results for the defined search strings. After the screening phases a total of 56 studies were selected as studies about agile methods tailoring. The selected studies were analyzed and the extracted data was classified into categories focusing on research aspects

(research type, empirical research and research validation) as well as technical aspects (agile methods covered and criteria for method tailoring). The results of the analysis and classification showed that this is a current and relevant research theme as the published studies have been growing over time and the studies are originated from multiple countries around the globe.

In general agile methods tailoring research can be considered mature since approximately 2/3 of the papers use empirical research methods. That is a fact that makes the claims of the studies stronger among fellow researchers but there are still papers claiming achievements with no supporting empirical evidence. The solution proposals dominated the research type aspect of the studies. That can be explained by the need for a model for agile method tailoring to be established and adopted by organizations. The validation of the studies was mainly focused on evaluation. The evaluations were conducted by questionnaires or by expert validation.

Method engineering has been the primary choice regarding method tailoring approaches combined with the usage of approaches capable of dealing with any agile method the organization has decided to adopt. The main tailoring criteria groups used for agile practice selection took into account internal environment and objectives criteria and generating reports or procedures as the research results.

The studies showed in majority agile method tailoring techniques that were independent of the agile method adopted by the organization. The techniques dependent of agile methods were based on Scrum and XP, the two most adopted agile methods according to existing research (Ahmed and Sidky, 2009; de Azevedo Santos et al., 2011; Glaiel et al., 2013; Jyothi and Rao, 2011; Moniruzzaman and Hossain, 2013; VersionOne, 2013). Method engineering was used as the method tailoring approach allowing the selection of agile practices based on certain criteria. The criteria utilized by the studies were mainly internal environment and objectives. The internal environment criteria used variables such as project type, communication, culture and management support. The objectives criteria used variables such as business goals, degree of innovation and complexity. A finding of this study was an extended classification for software development tailoring criteria proposed by the authors and based on the work of Kalus and Kuhrmann (Kalus and Kuhrmann, 2013) and described on Table 11. This extended criteria was proposed to

provide a better classification for the analyzed articles and clarify which criteria has been used on the analyzed research papers.

Analysis of previous research work showed that a comprehensive systematic review covering agile methods tailoring didn't exist yet. The contributions of this study are a systematic outline of agile methods tailoring research summarizing the research and technical aspects found on the literature and the indication of research paths that can be followed by researchers when proposing new agile method tailoring approaches and also provide initial guidance for organizations on how start their tailored agile method adoption to achieve minimal disruption on their existing software development method context. Currently we are working on a model for agile methods tailoring based on the results of the literature review and the relationship between criteria for agile method tailoring and agile practices selected for adoption by the organizations. Future research should explore method tailoring approaches to understand the articles classified as unclear and check if there is a need for a new classification of method tailoring approaches or if there is a lack of awareness of the method tailoring approaches on the agile methods tailoring research community.

Appendix A. List of analyzed papers

The final list of analyzed papers on the SLR is summarized on Table A.21.

Table A.21

Final list of analyzed papers on the SLR.

Database	Articles
ACM Digital Library	Mnkandla (2005), Coelho et al. (2005), Mnkandla et al. (2005), Pikkarainen and Salo (2006), Taylor et al. (2006), Fitzgerald et al. (2006), Taylor et al. (2008), Mirakhorli et al. (2008), Esfahani et al. (2010), Conboy and Fitzgerald (2010), Soundararajan and Arthur (2011), Hossain et al. (2011), Ayed et al. (2012), Bass (2013), Ayed et al. (2014)
IEEE Xplore	Keenan (2004), Little (2005), Meng et al. (2007), Mnkandla and Dwolatzky (2007), Qumer and Henderson-Sellers (2007), Hussain et al. (2008), El-Said et al. (2009), de Souza Mariz et al. (2010), Qumer and Henderson-Sellers (2010), Hayata and Han (2011), Wang (2011), Soundararajan et al. (2012), Escobar-Sarmiento and Linares-Vasquez (2012), Ortega et al. (2012), Jonsson et al. (2012), Bhalerao and Ingle (2012), Bass (2012), Murugaiyan and Balaji (2012), Mukhtar et al. (2013), Soundararajan et al. (2013), McHugh et al. (2013), Rakshith and Patil (2013), Khan and Beg (2013), Hamed and Abushama (2013), Tripp and Armstrong (2014), Sultana et al. (2014), Rodriguez et al. (2014)
Science Direct	Qumer and Henderson-Sellers (2008), Williams (2010), van Waardenburg and van Vliet (2013)
Springer	Pikkarainen et al. (2005), Sidky et al. (2007), Krasteva and Ilieva (2010), Rohunen et al. (2010), Krasteva et al. (2010), Mikulenas and Kapocius (2011), Papatheocharous and Andreou (2013), Papatheocharous et al. (2013)
Wiley Online	Kruchten (2013), McHugh et al. (2014), Machado et al. (2014)

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